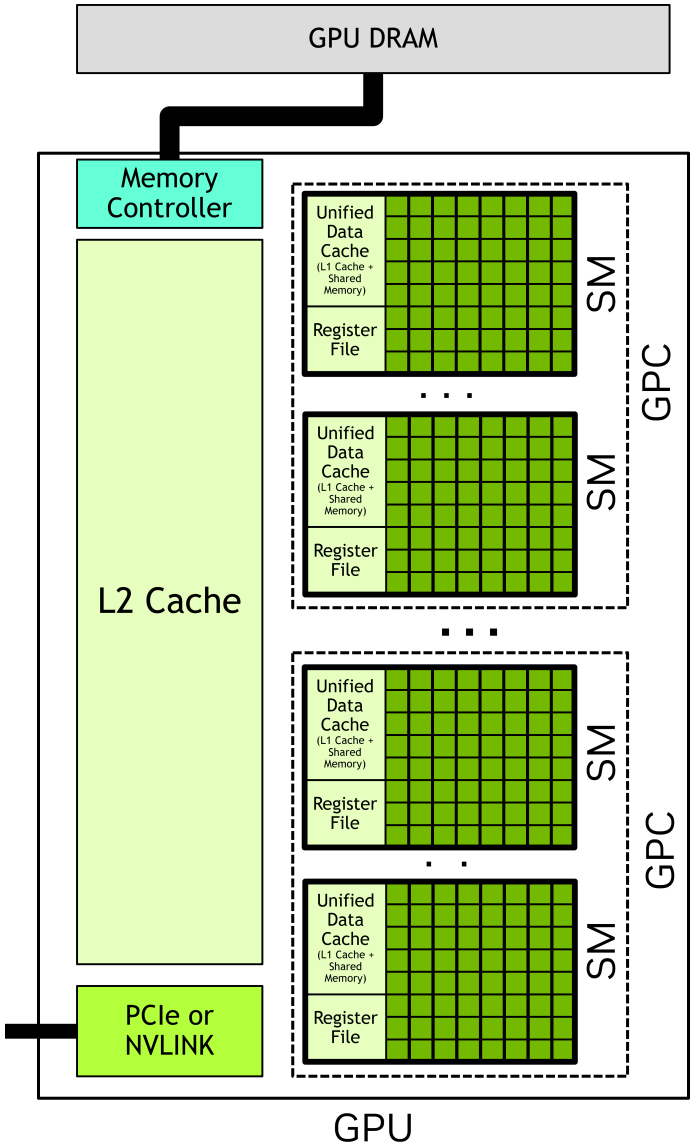
Matrix Multiplication on GPUs

Max Koch

GPU Architecture Recap

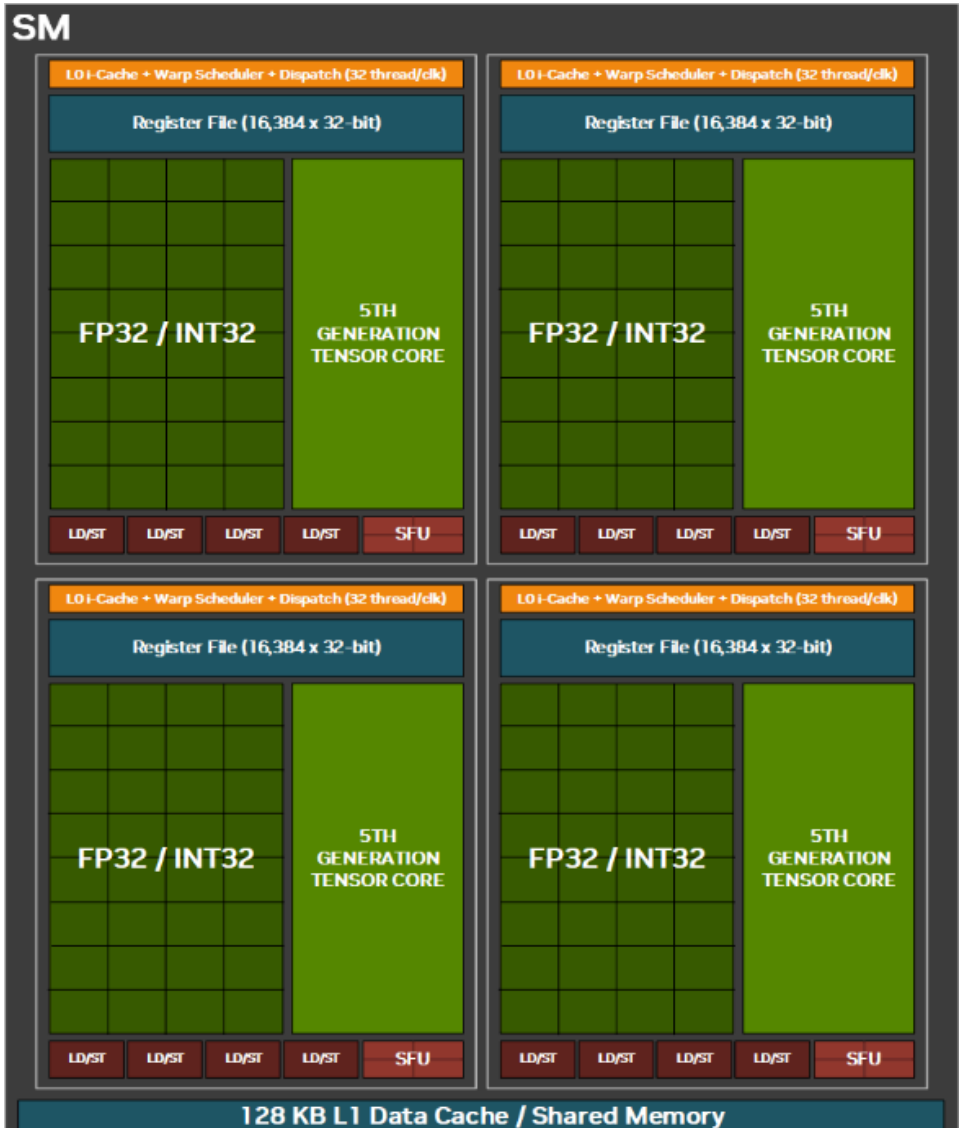
- GPUs have an array of **Streaming Multiprocessors (SMs)**
- Each SM is an independent parallel processor
- All SMs share an **L2 cache**
- Off-chip **HBM** (High Bandwidth Memory) — high bandwidth, high latency
- CPU and GPU communicate over PCIe or NVLink



GPU system with SM array, shared L2, and HBM
Adapted from: NVIDIA — CUDA Programming Guide

GPU Architecture Recap

- Each SM contains:
  - **CUDA cores**: contain ALUs for scalar integer and floating-point ops
  - **Tensor Cores**: hardware matrix multiply-accumulate (MMA)
  - **Shared memory / L1 cache**: fast on-chip scratchpad memory
  - **Register file**: large, per-thread
  - **Warp schedulers**: dispatch groups of 32 threads, fast context switching
- Warps execute in **SIMT**: single instruction, multiple threads

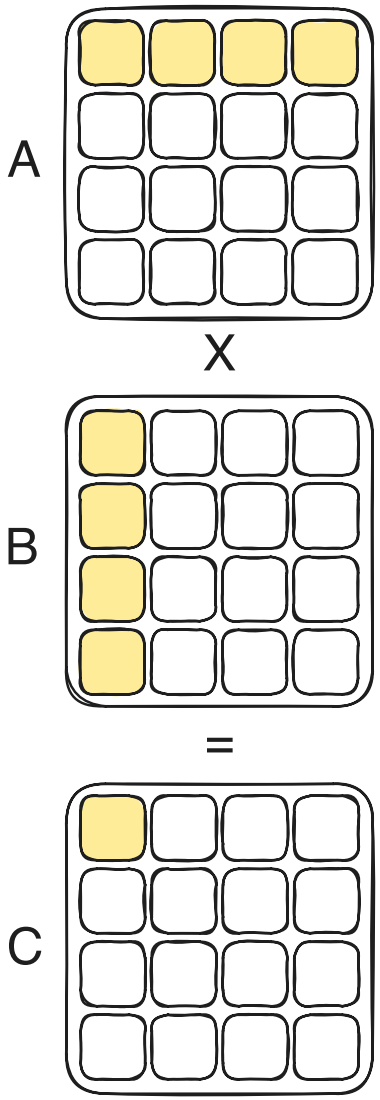


The Blackwell Streaming Multiprocessor (SM)
Adapted from: NVIDIA — NVIDIA RTX Blackwell GPU Architecture

Matrix Multiplication

- One of the most important kernels in deep learning
- Used for:
  - Model training and inference
  - Rendering graphics
  - Scientific computing
  - …

$$C = A \times B$$



Matrix Multiplication with CUDA Cores

- Each thread is responsible for computing one (or multiple) output elements of C
- For each output element, the thread loops over the reduction dimension and accumulates the products of corresponding elements from A and B

- **Problems:**
  - Different threads have to load the same elements of A and B to their respective registers
  - Each thread has to load, compute, and store elements in each iteration of the reduction loop
  - Support for *mixed precision* is limited (would require manual dtype promotion)

Tensor Cores solve these problems.

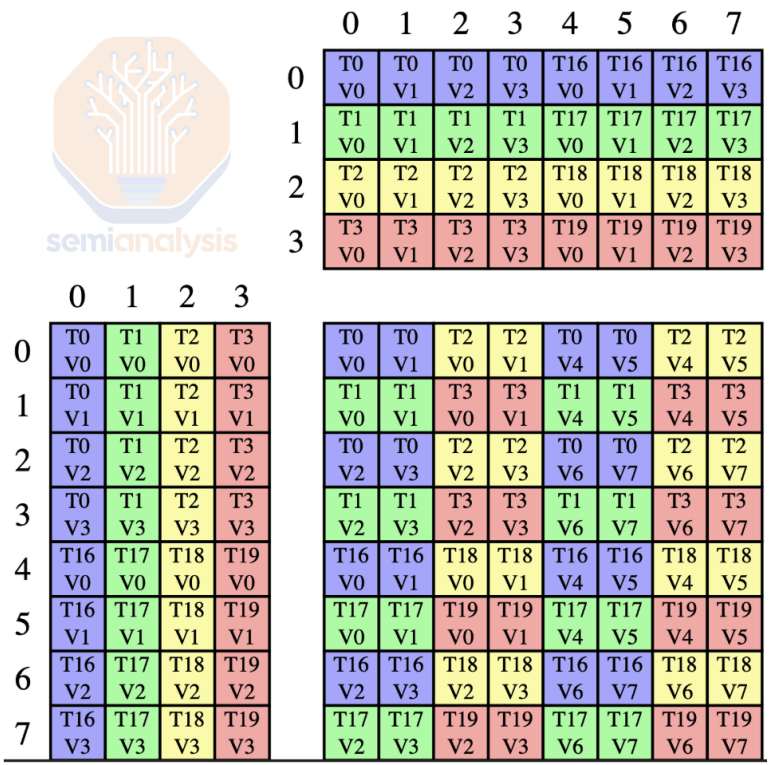
Tensor Cores

- Specialized hardware units introduced with **Volta** (2017)
- Execute matrix multiply-accumulate $D = A \times B + C$ in a single instruction
- Operate on **fixed tile shapes**
- Support **mixed precision** with automatic accumulation in higher precision
- Programming perspective: Multiple threads collaborate on one matrix multiplication

Blackwell Tensor Core — FP4
From: NVIDIA — NVIDIA RTX Blackwell GPU Architecture

Tensor Cores: Element Assignment to Registers (Volta)

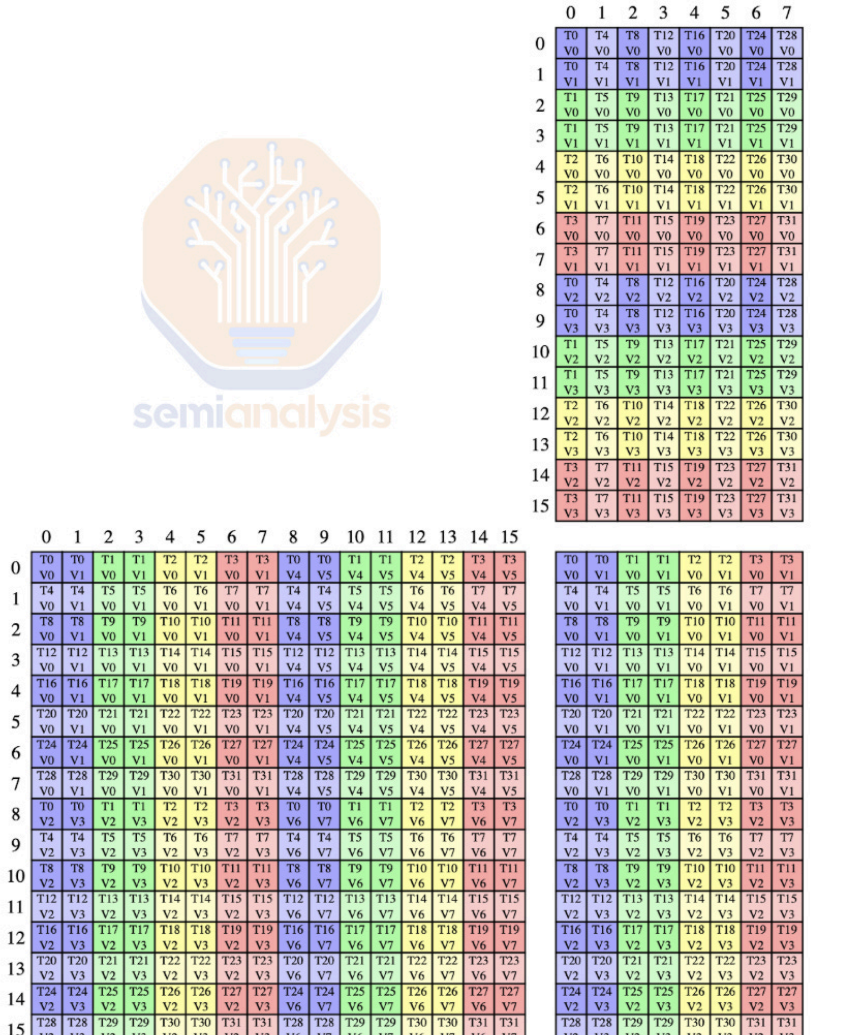
- Each thread is responsible for loading a specific subset of elements of A and B into registers



Register layout on Volta
From: SemiAnalysis — NVIDIA Tensor Core Evolution: From Volta To Blackwell

Tensor Cores: Ampere

- Tensor Cores operate on larger tiles compared to Volta
- Thread assignments differ between Volta and Ampere
- Ampere's Tensor Cores operate at the full-warp level



Register layout on Ampere
From: SemiAnalysis — NVIDIA Tensor Core Evolution: From Volta To Blackwell

Tensor Cores: Matrix Sizes

- Different architectures support different tile sizes for A, B, and C

Tensor Core Size Increases			
Architecture	**Tensor Core FLOP/Cycle/SM**	**Max MMA Shape**	**Max FLOPs per PTX Instruction (2 * m * n * k)**
Volta	F16: 1024	m8n8k4	512
Ampere	F16: 2048	m16n8k16	4096
Hopper	F16: 4096 F8: 8192	**Warpgroup Level** **F16:** m64n256k16 **F8:** m64n256k64	**Warpgroup Level** **F16:** 524,288 **F8:** 2,097,152
Blackwell	F16: 8192 F8: 16384 F4: 32768	**2 SM** **F16:** m256n256k16 **F8:** m256n256k32 **F4:** m256n256k96	**2 SM** **F16:** 2,097,152 **F8:** 4,191,304 **F4:** 12,582,912

From: SemiAnalysis — NVIDIA Tensor Core Evolution: From Volta To Blackwell

Tensor Cores: Memory Levels

- Different architectures expect matrix elements to reside in different memory locations

Matrix Memory Residency across NVIDIA GPU Architectures			
Architecture	**Matrix A**	**Matrix B**	**Matrix D**
Volta	RF	RF	RF
Ampere	RF	RF	RF
Hopper	RF / SMEM	SMEM	RF
Blackwell	SMEM / TMEM	SMEM	TMEM

From: SemiAnalysis — NVIDIA Tensor Core Evolution: From Volta To Blackwell

Using Tensor Cores in cuTile

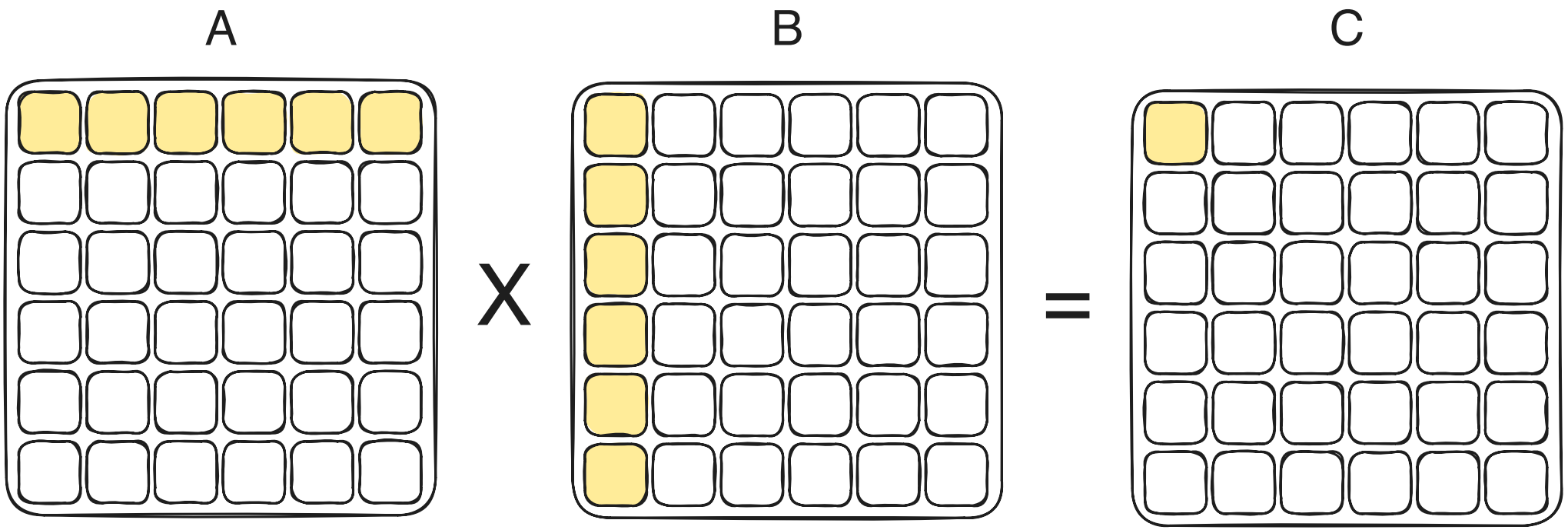
- cuTile provides a high-level API for programming Tensor Cores: `ct.mma`

```python
import cuda.tile as ct
import torch

@ct.kernel
def kernel():
    x = ct.ones((2, 4), dtype=ct.float16)
    y = ct.ones((4, 2), dtype=ct.float16)
    acc = ct.full((2, 2), 10.0, dtype=ct.float32)
    # (x @ y) + acc: each element = 1*4 + 10 = 14
    print(f"{ct.mma(x, y, acc):.1f}")


torch.cuda.init()
ct.launch(torch.cuda.current_stream(), (1,), kernel, ())
torch.cuda.synchronize()
```

Adapted from: NVIDIA — cuTile Python `cuda.tile.mma`

Tiled Matrix Multiplication

- For each output block of C, a block row of A has to be multiplied with a block column of B



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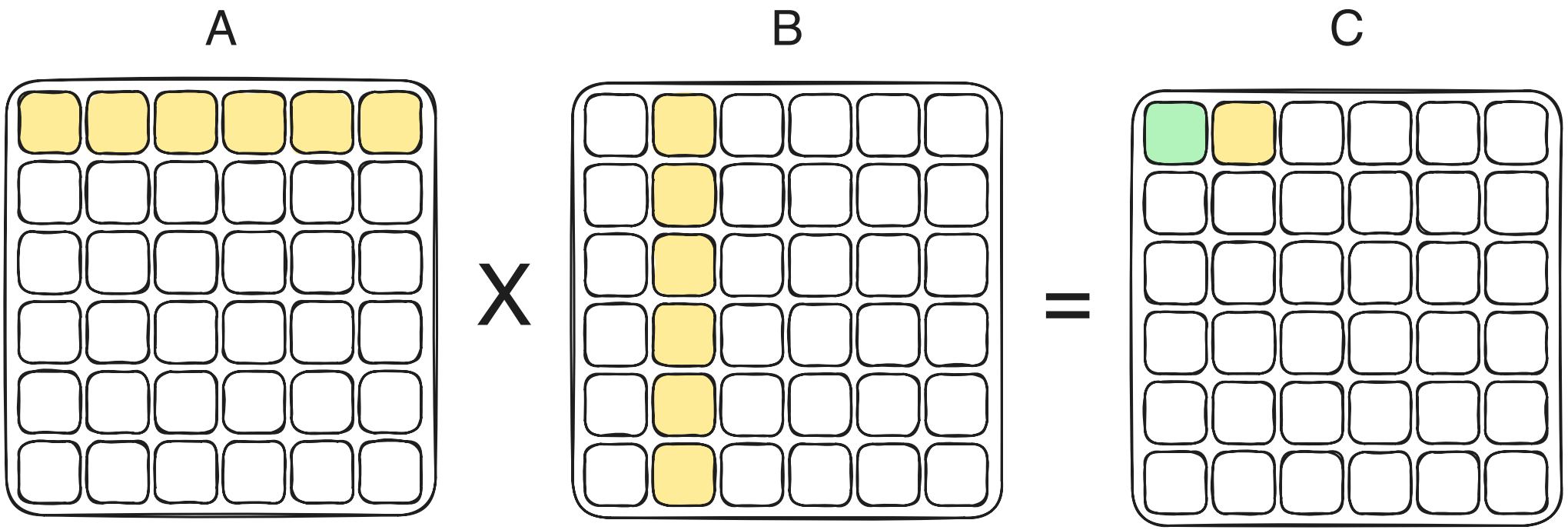

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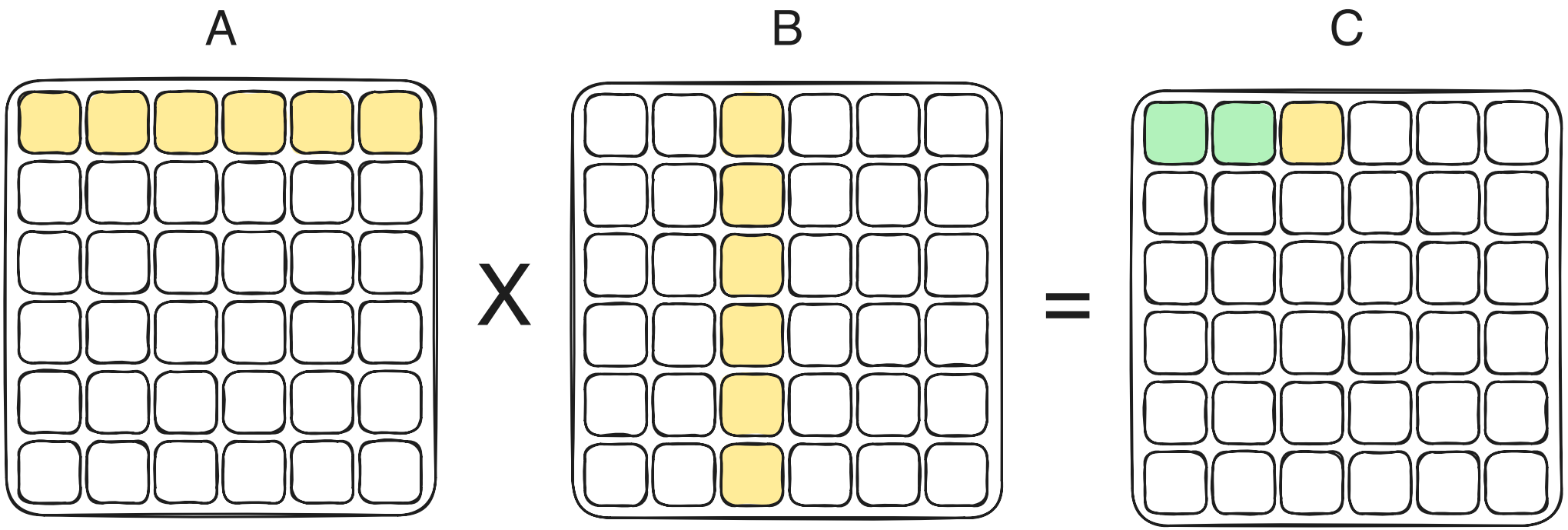

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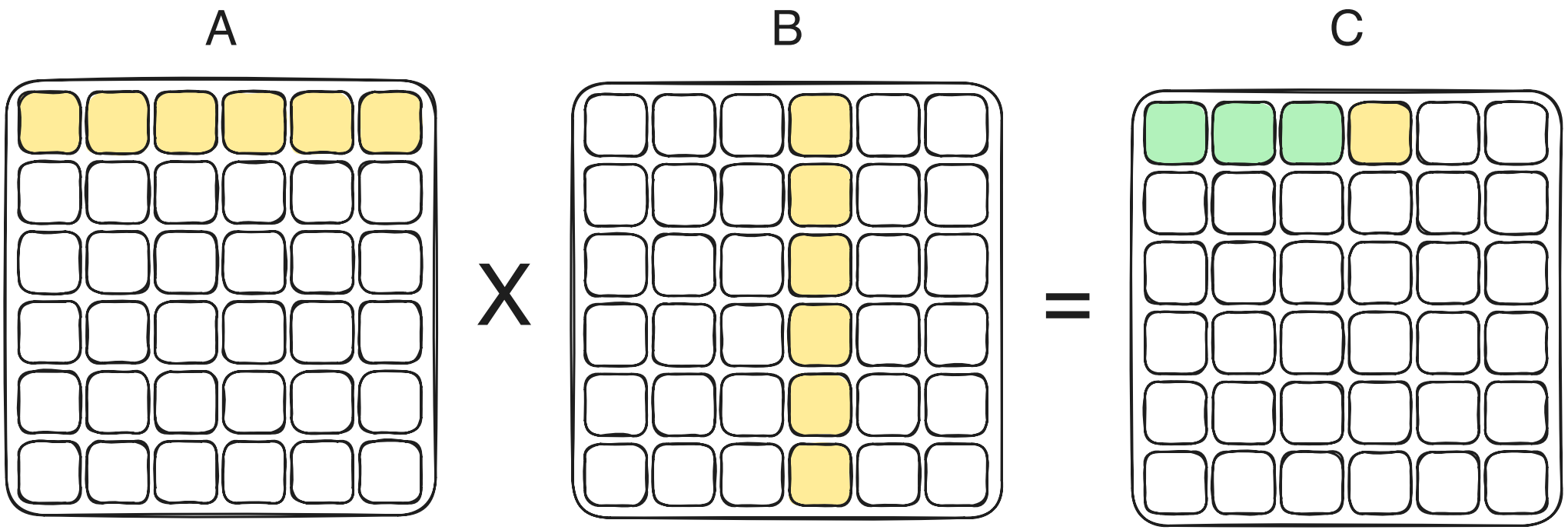

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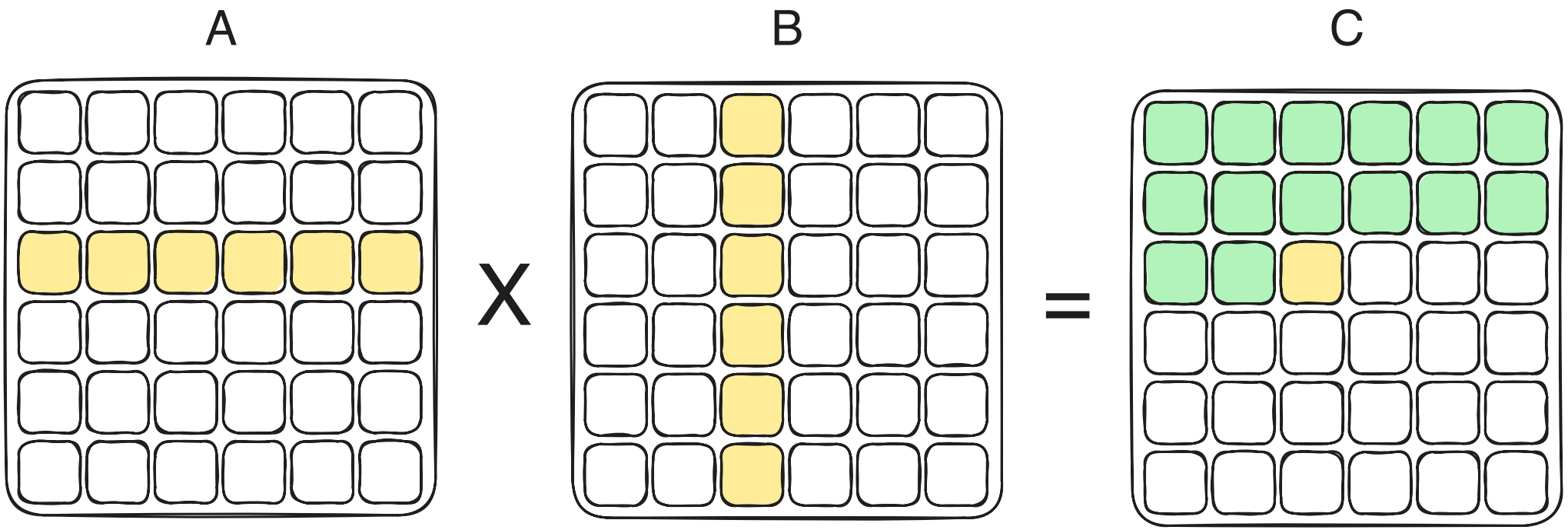

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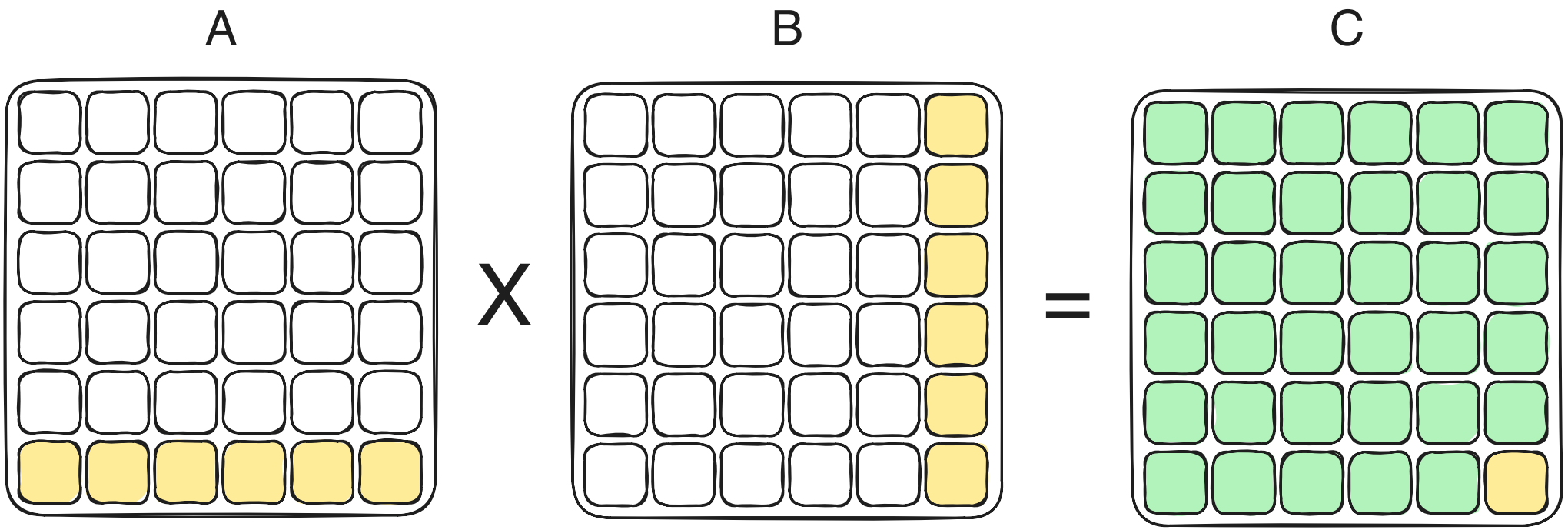

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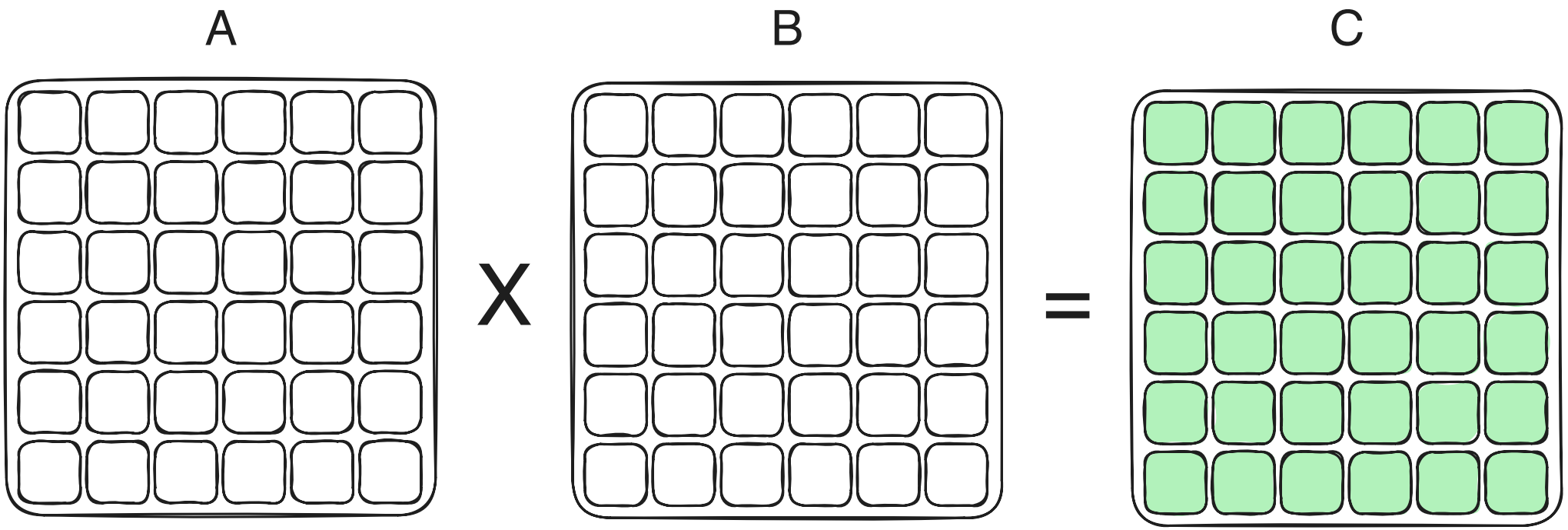
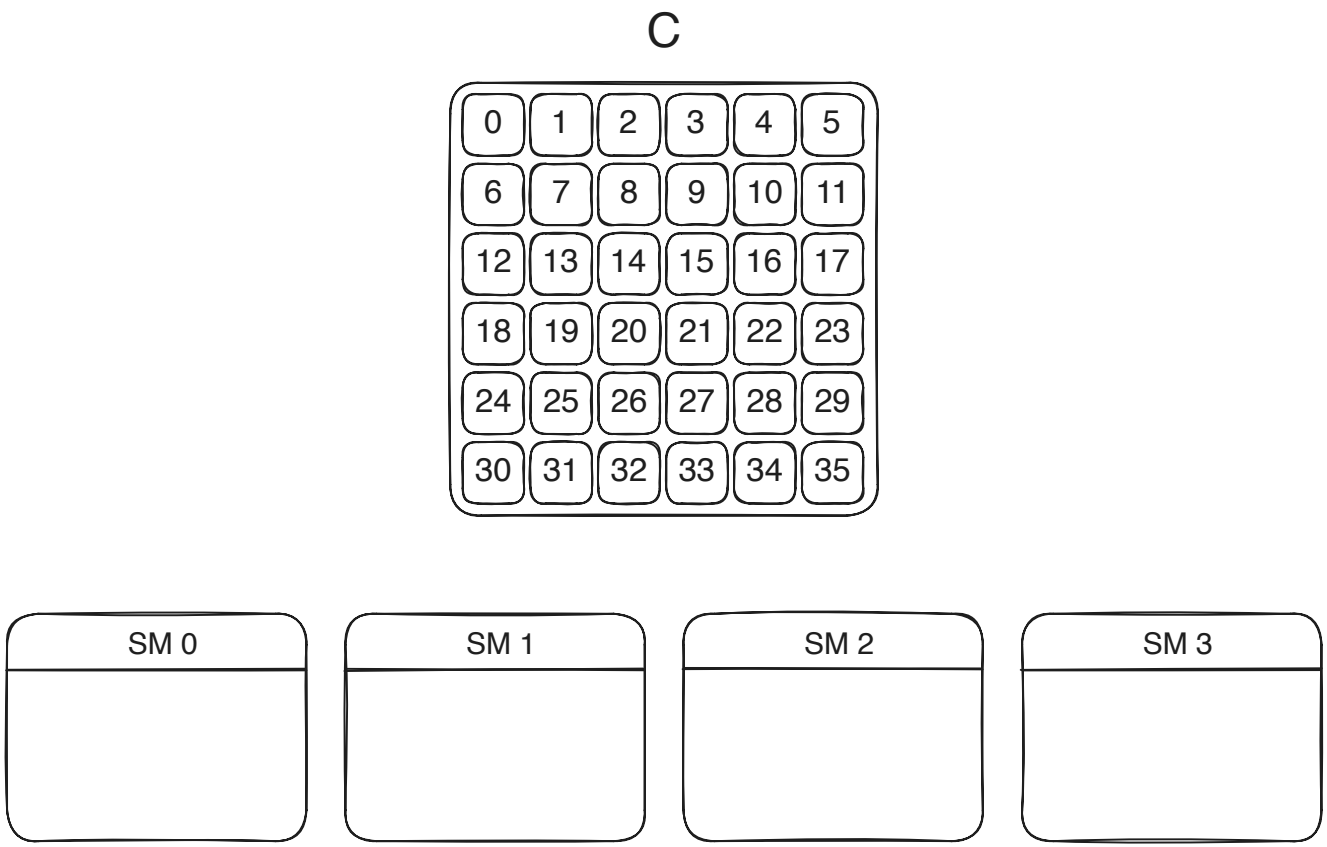


Assignment of Blocks to SMs

- Each output tile of matrix C can be computed in parallel



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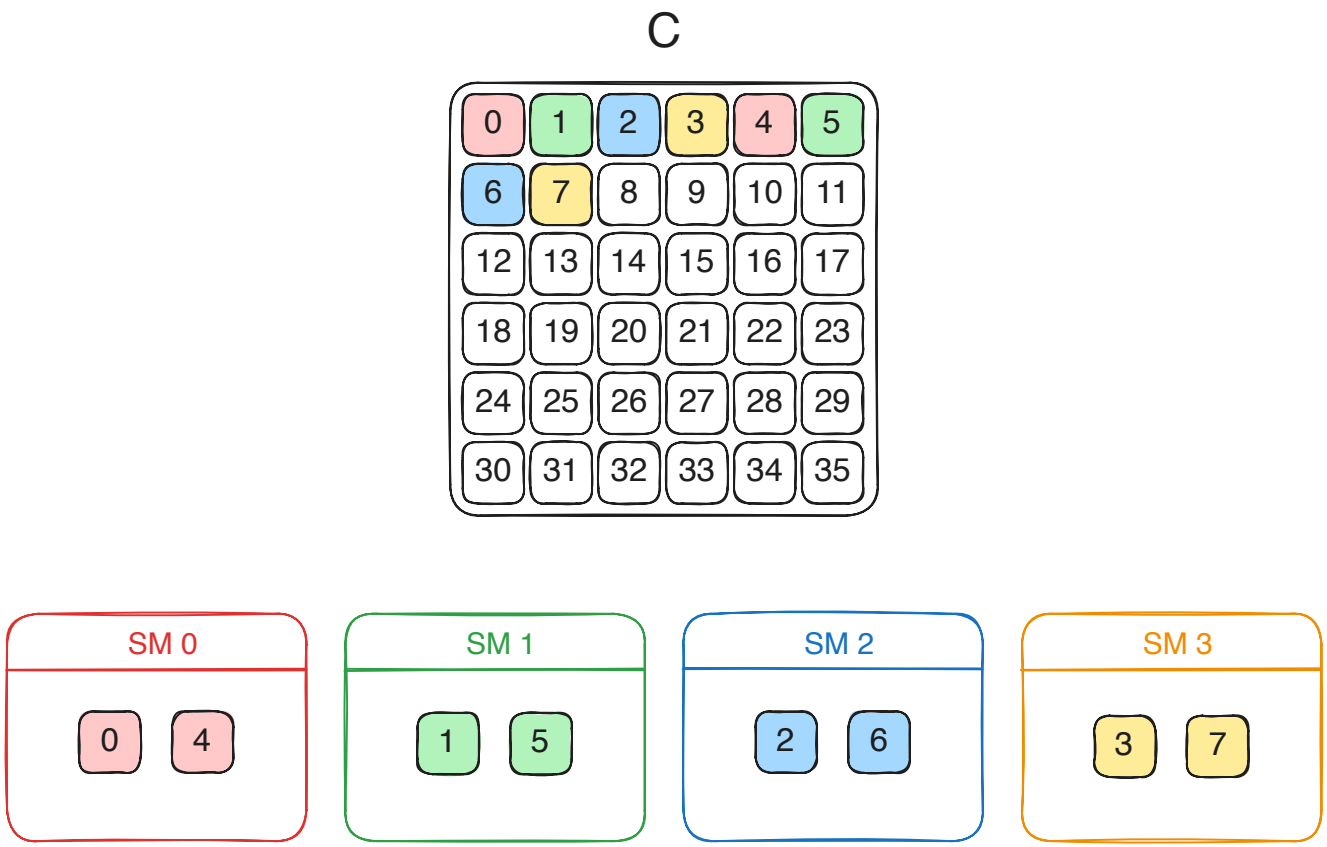

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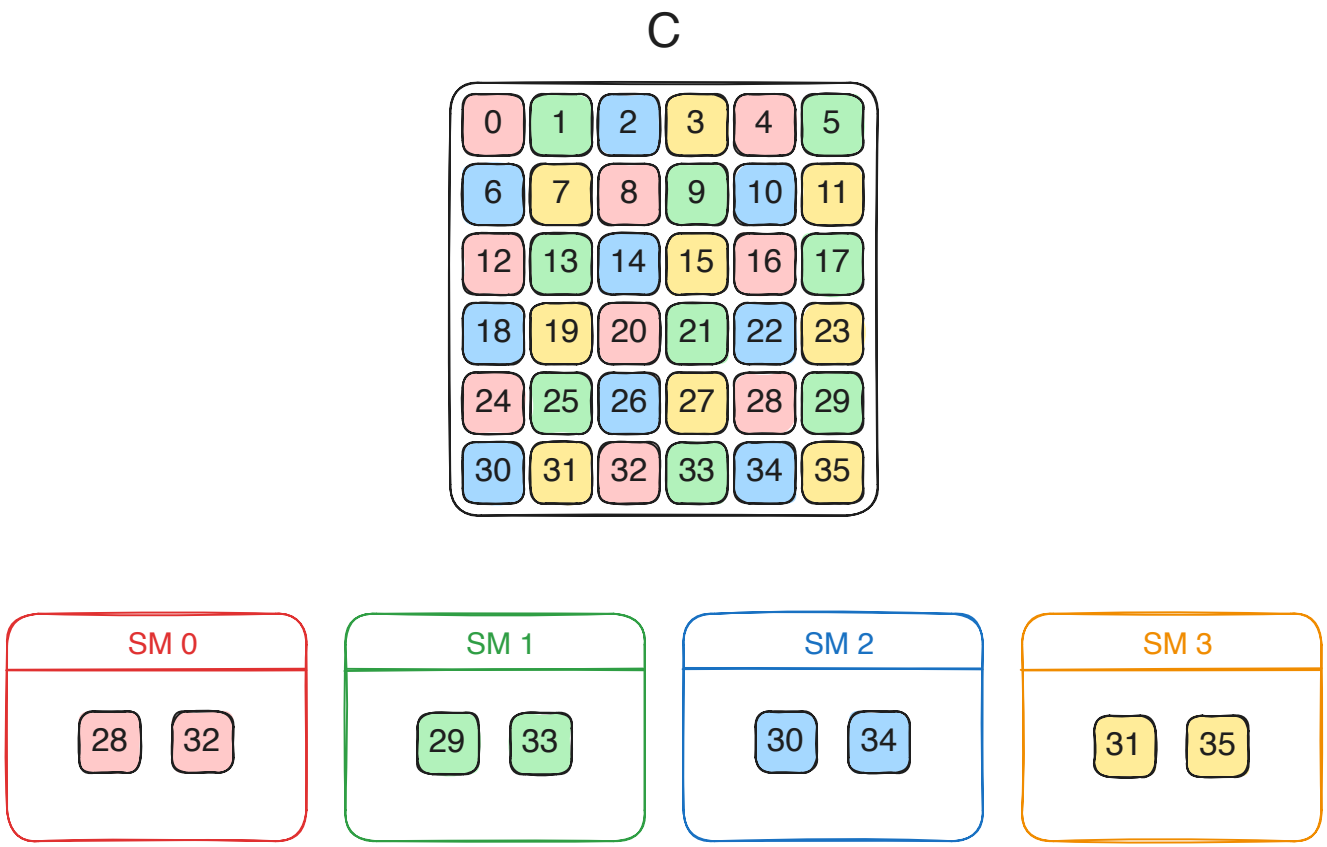
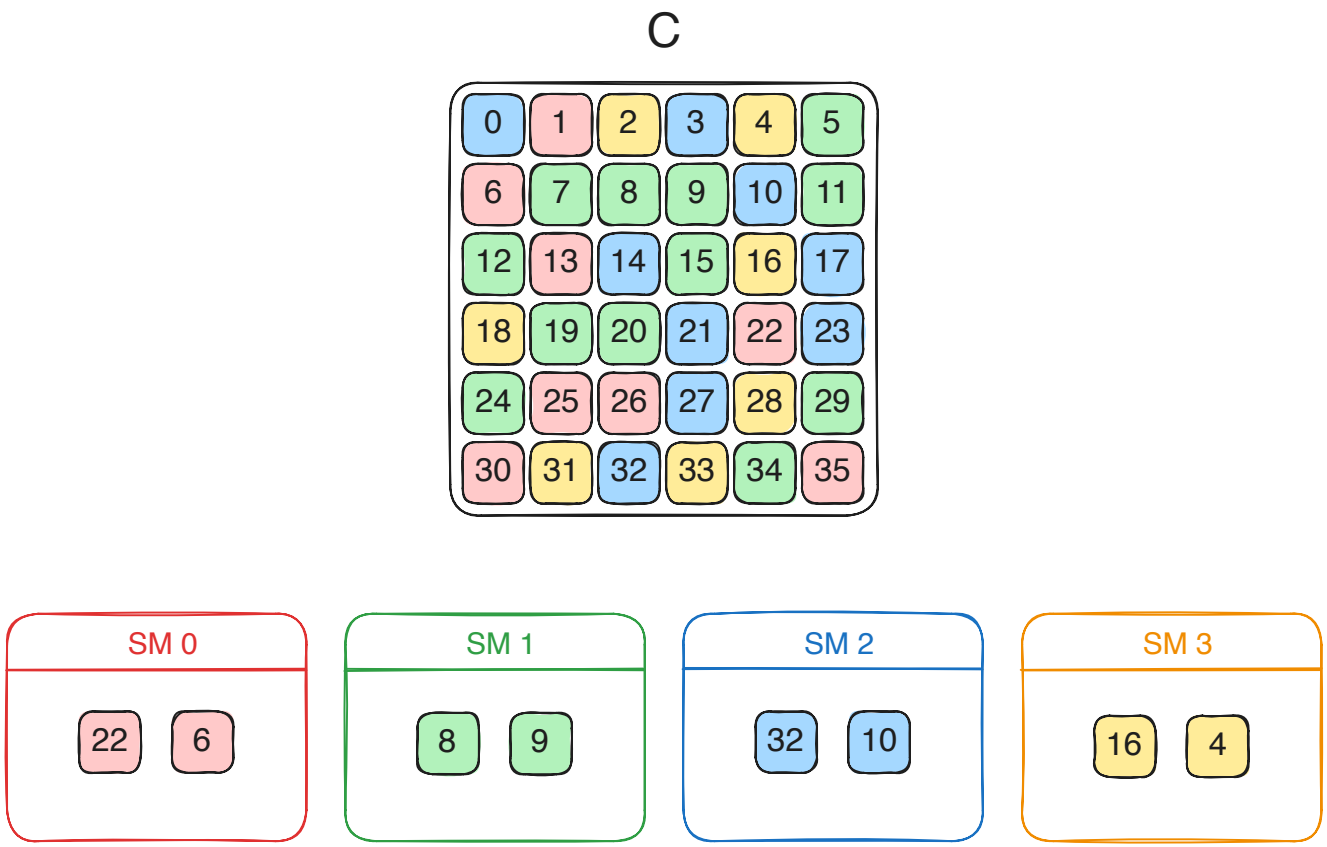


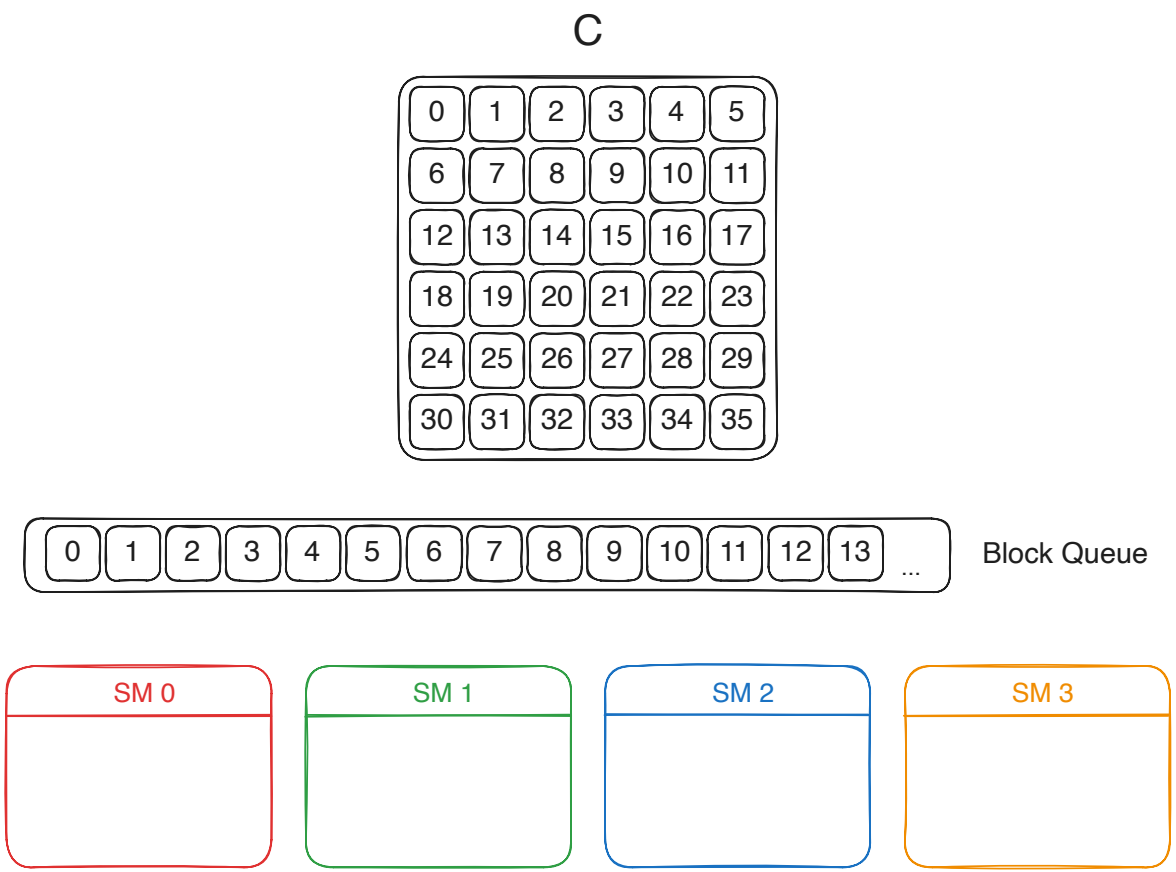
Assignment of Blocks to SMs

- There are no guarantees on the order in which blocks are assigned to SMs



Scheduling of Blocks to SMs

- In practice, the hardware scheduler assigns blocks ordered by their block ID



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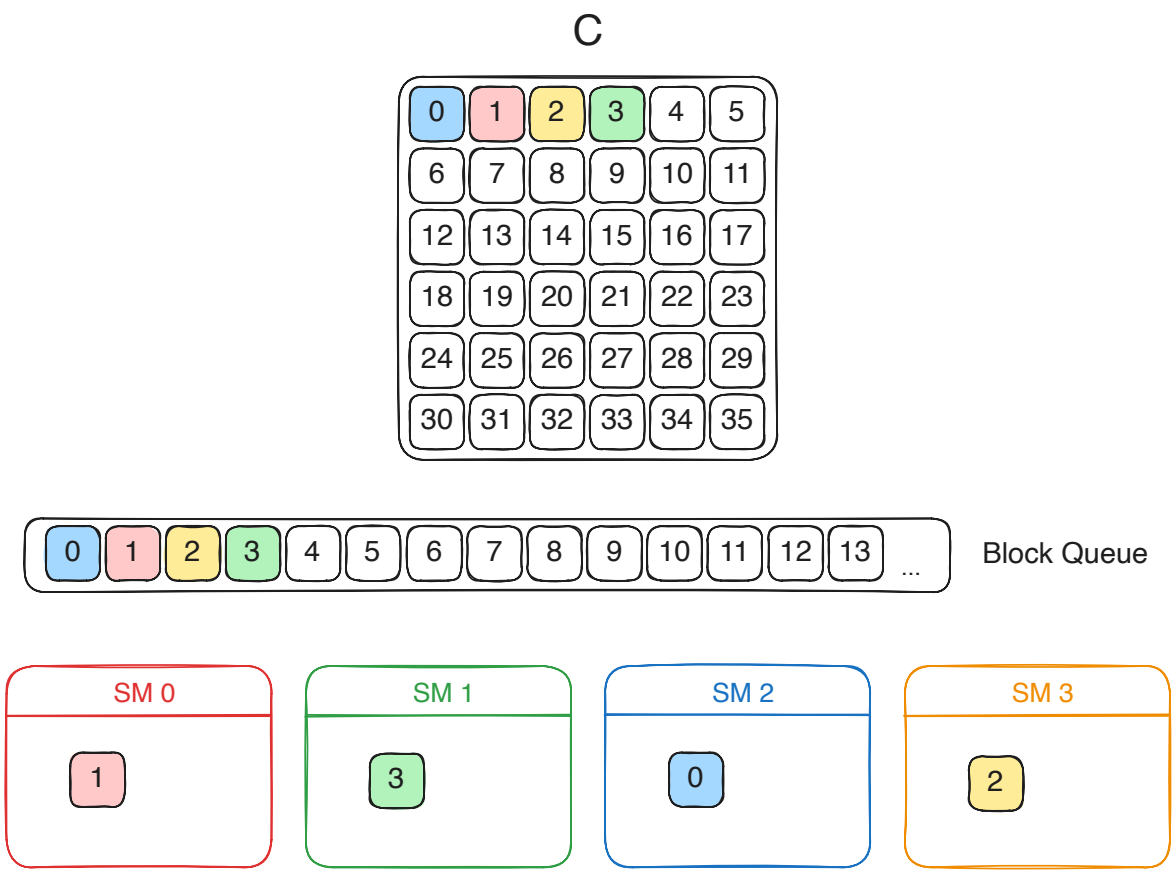

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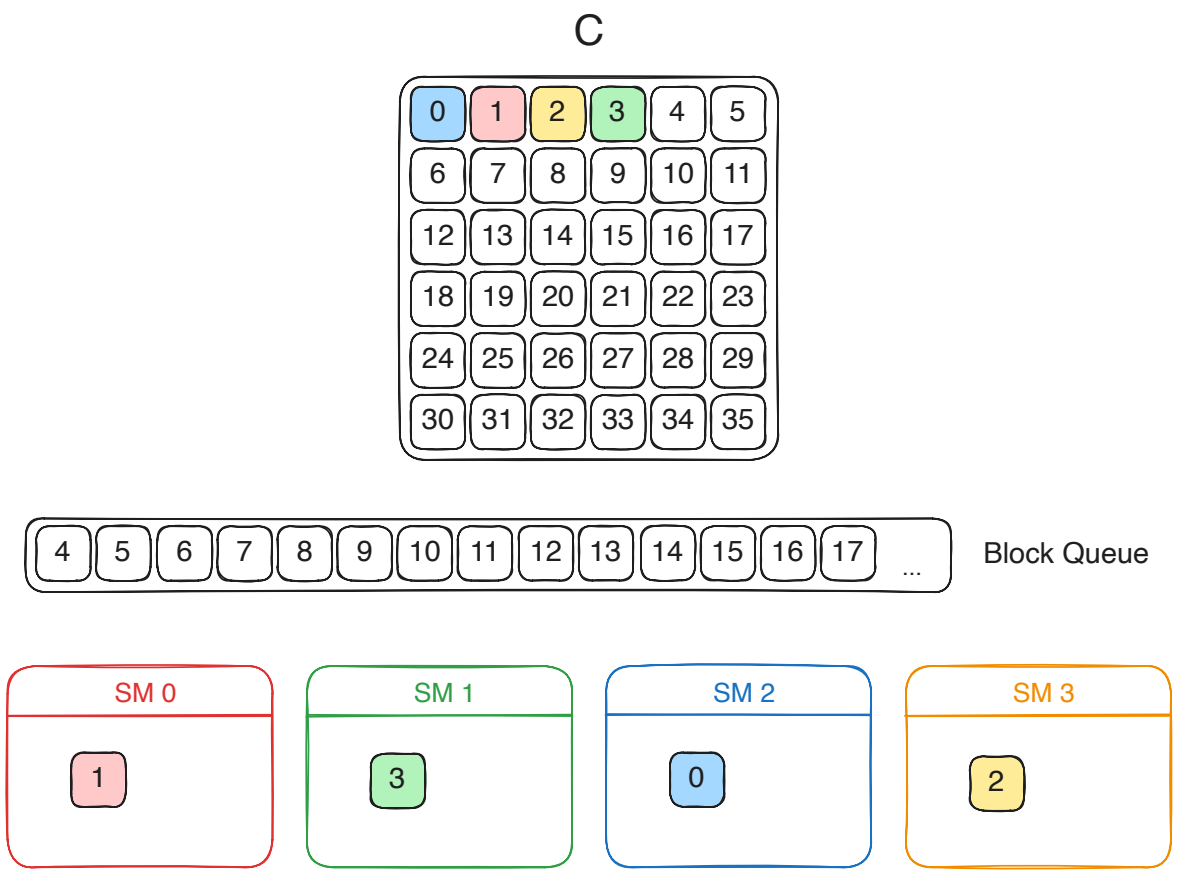

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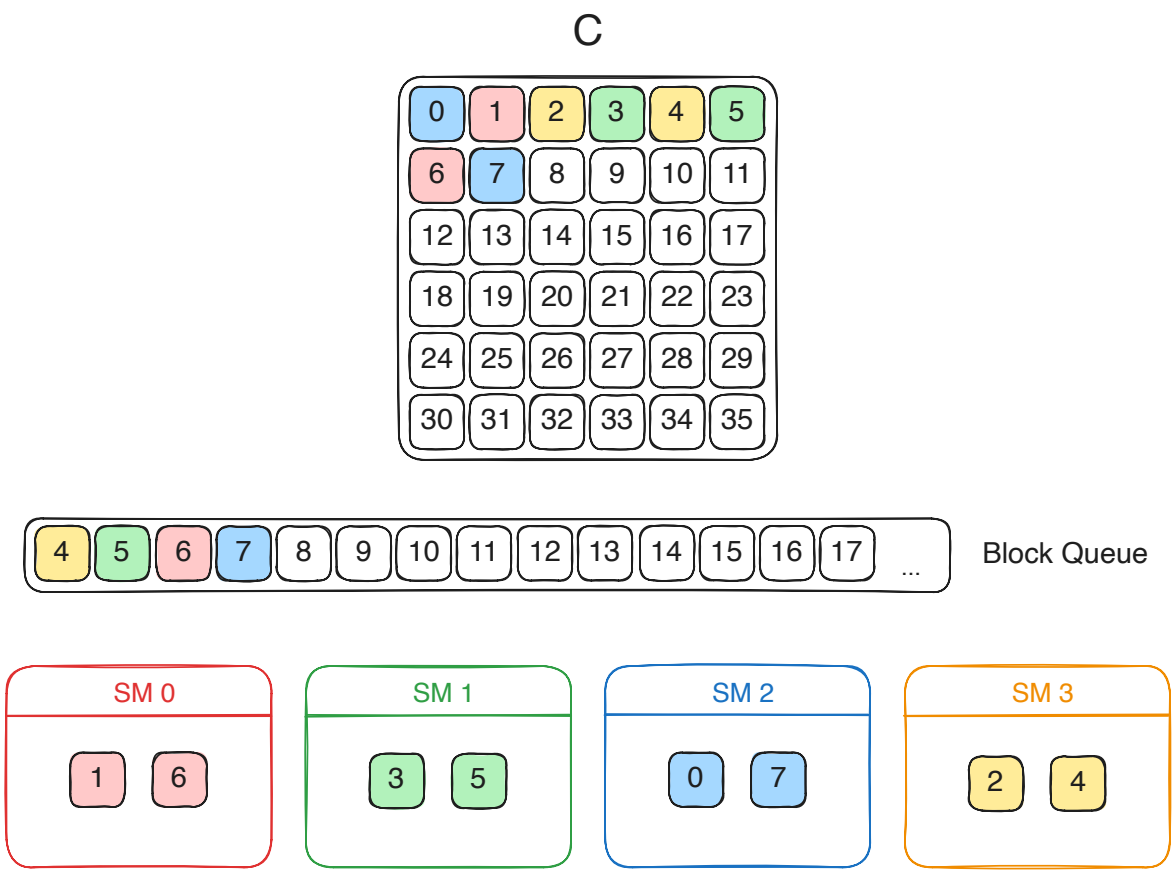

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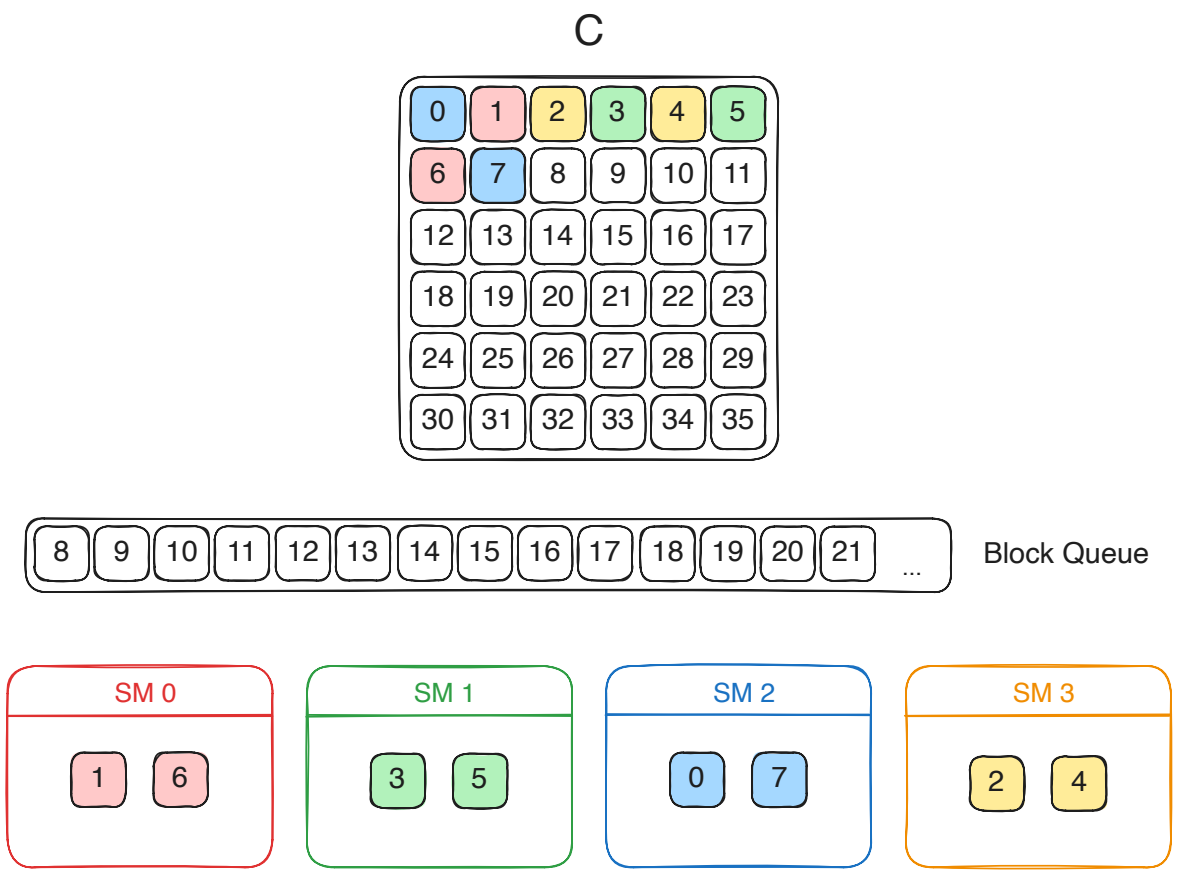

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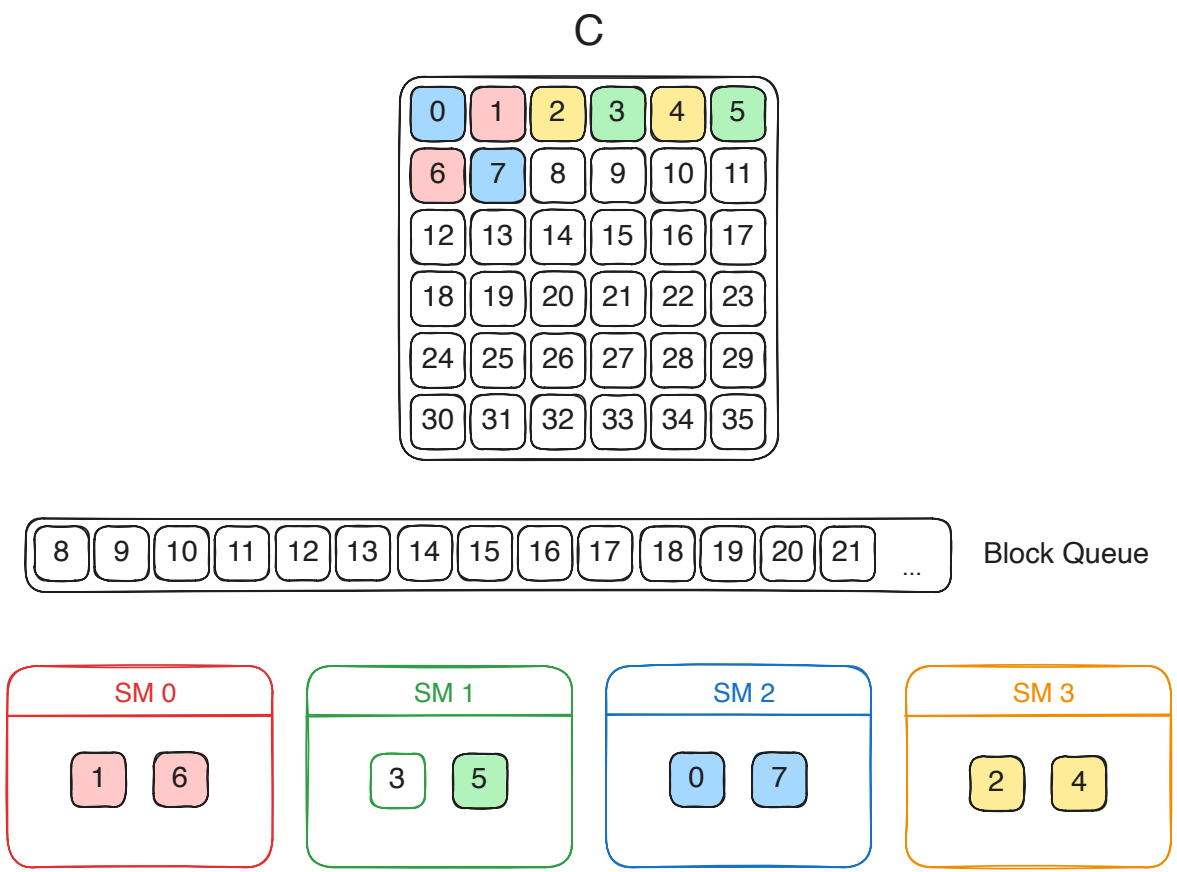

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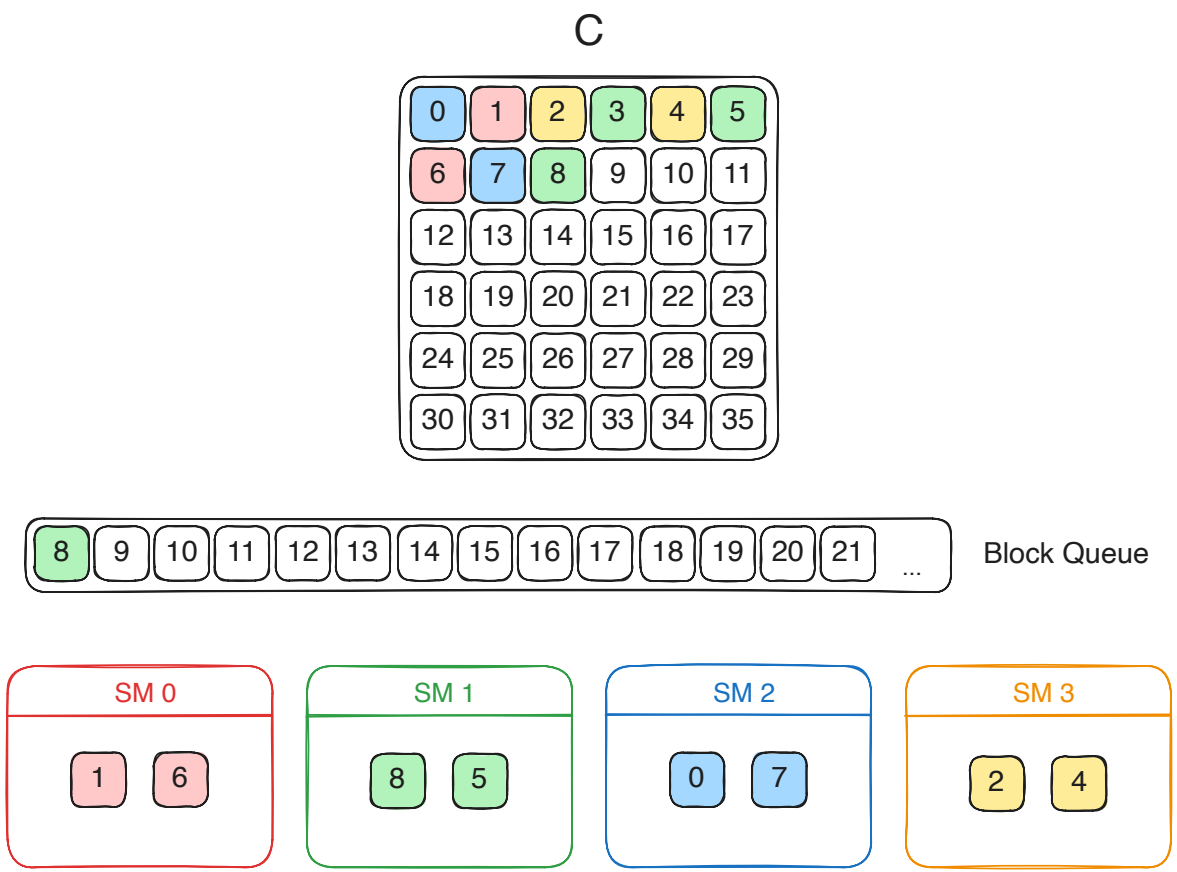

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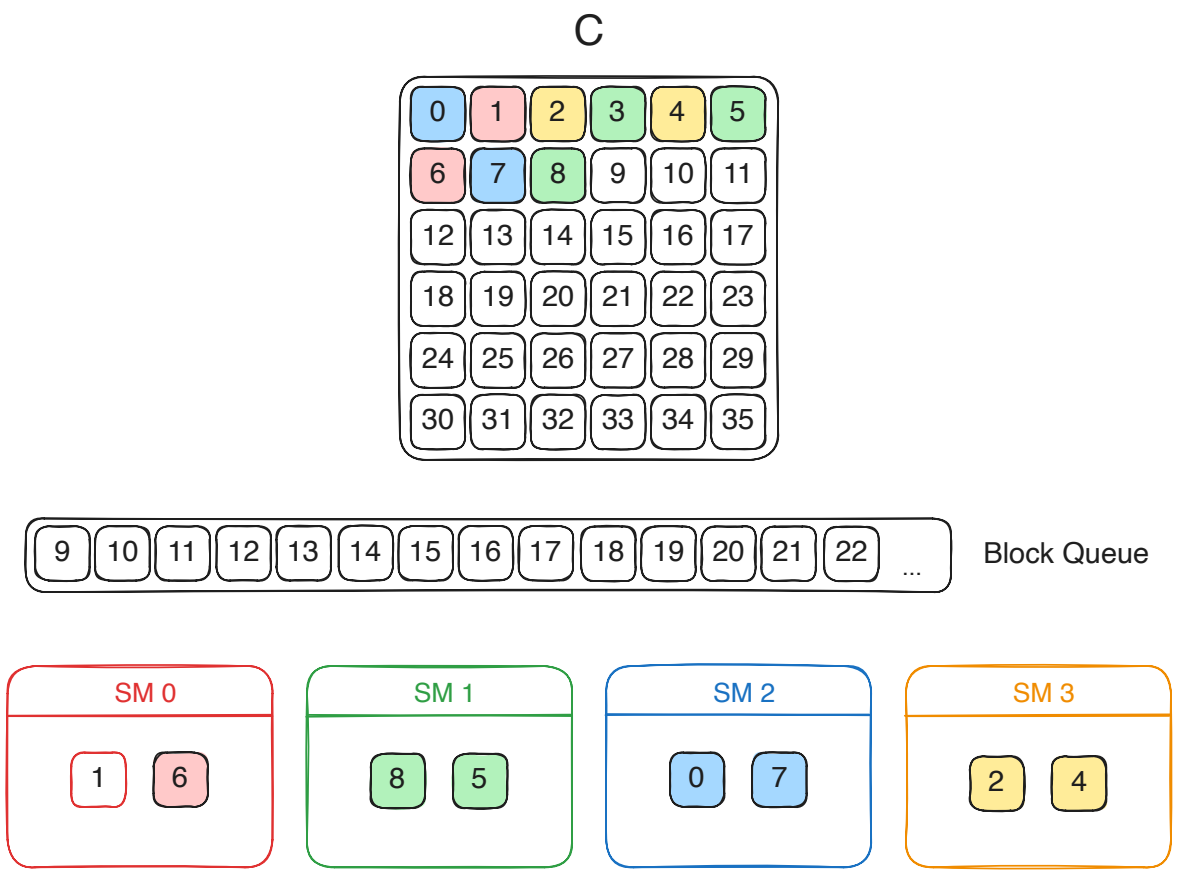

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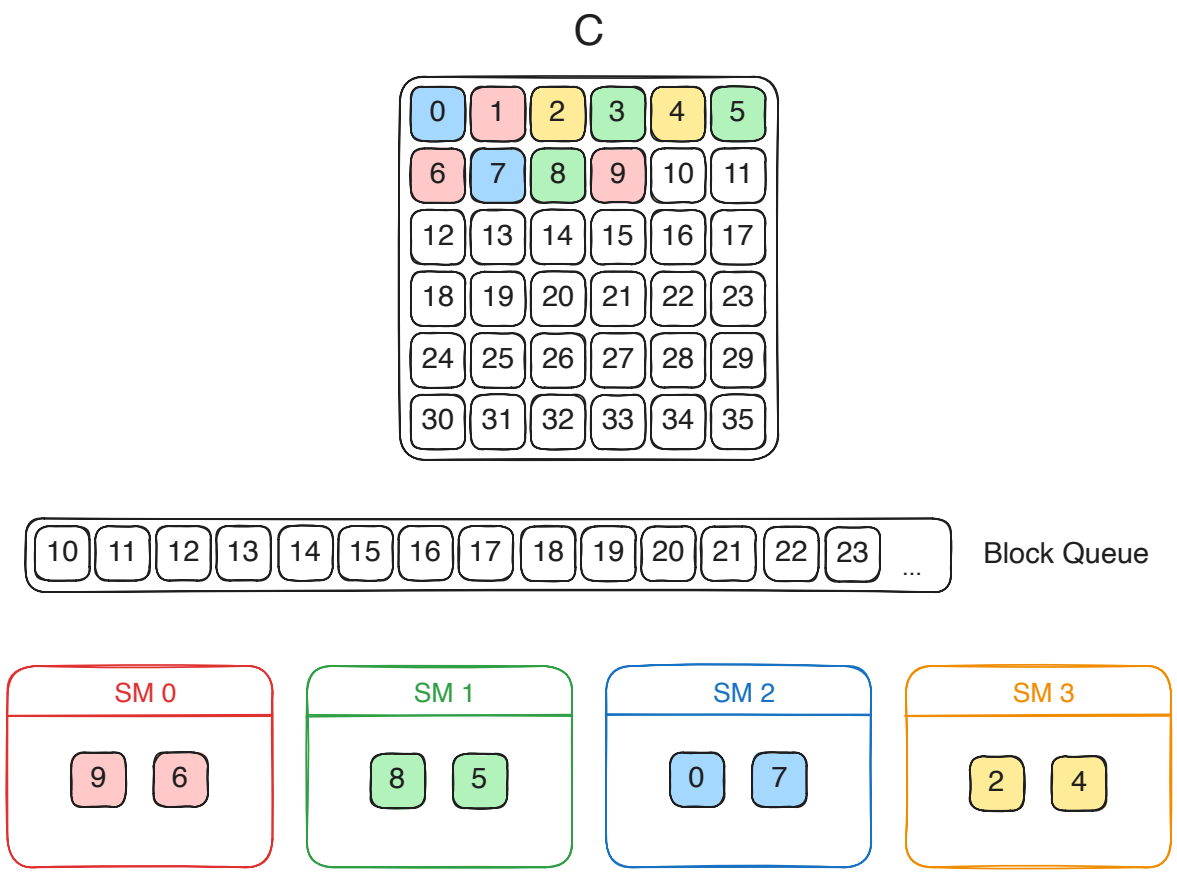

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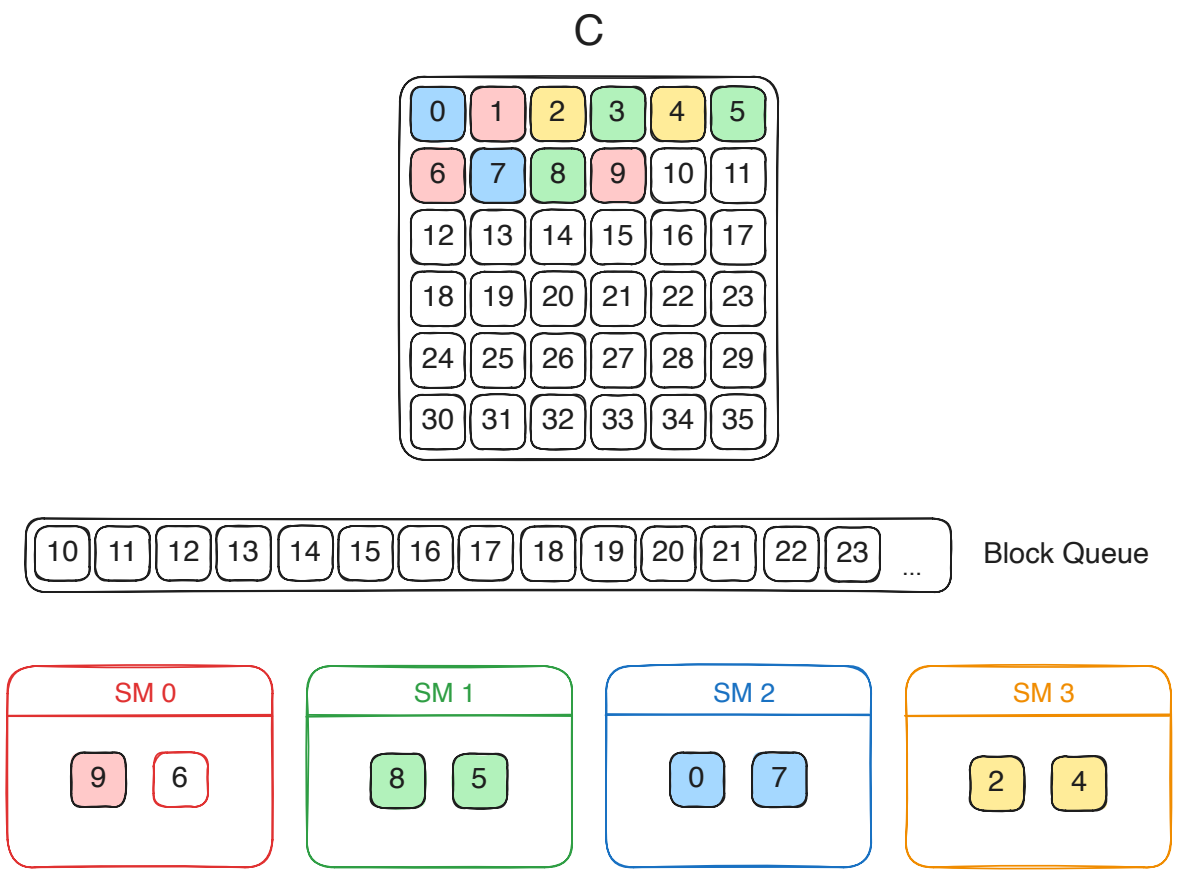

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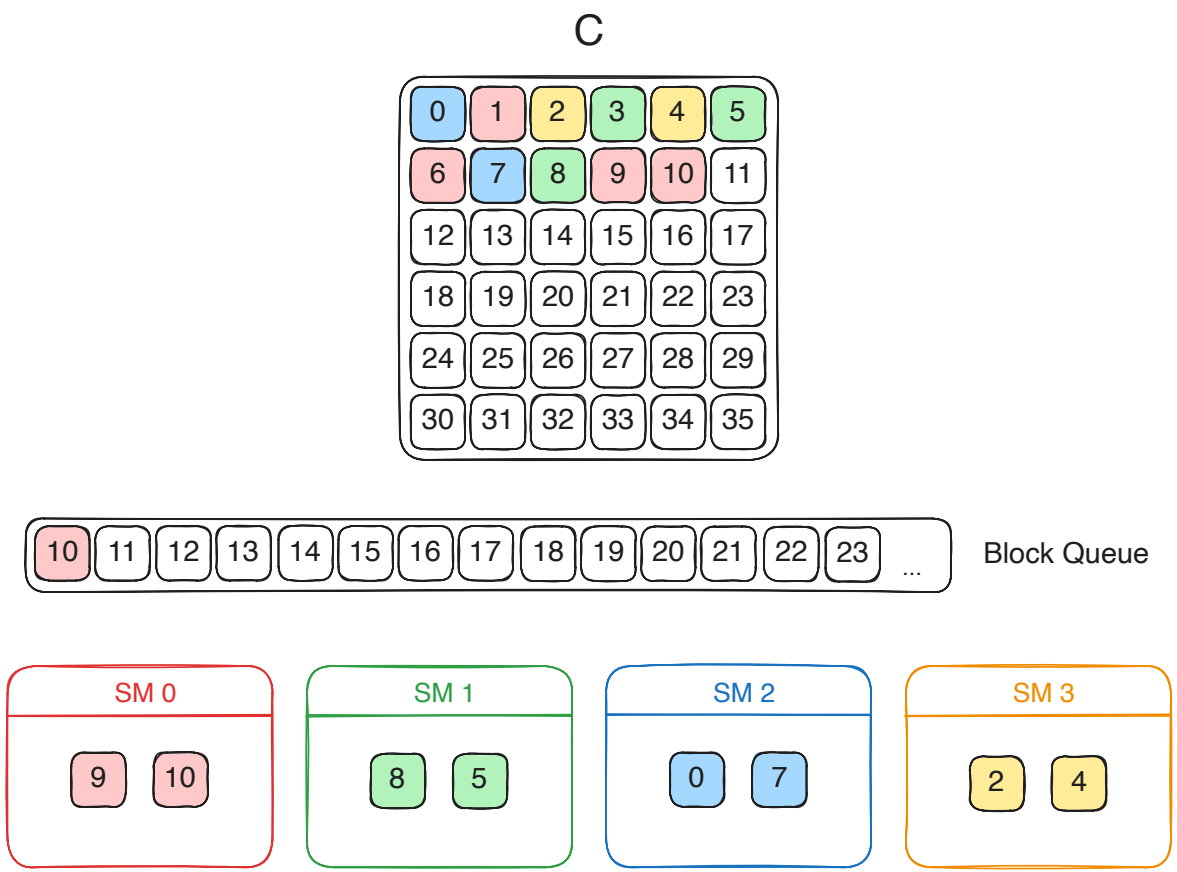
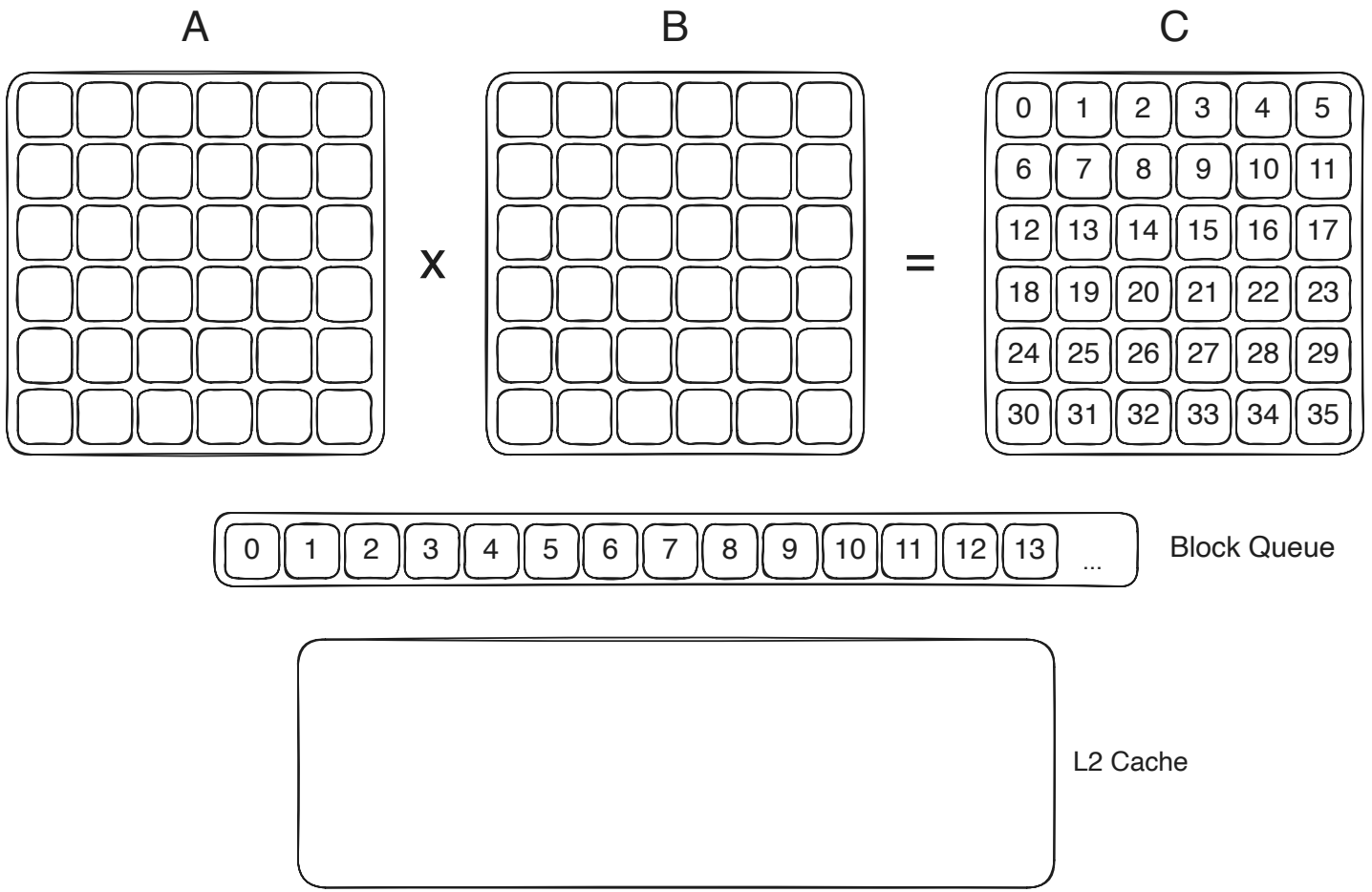


Effects of Block Scheduling on Cache Reuse



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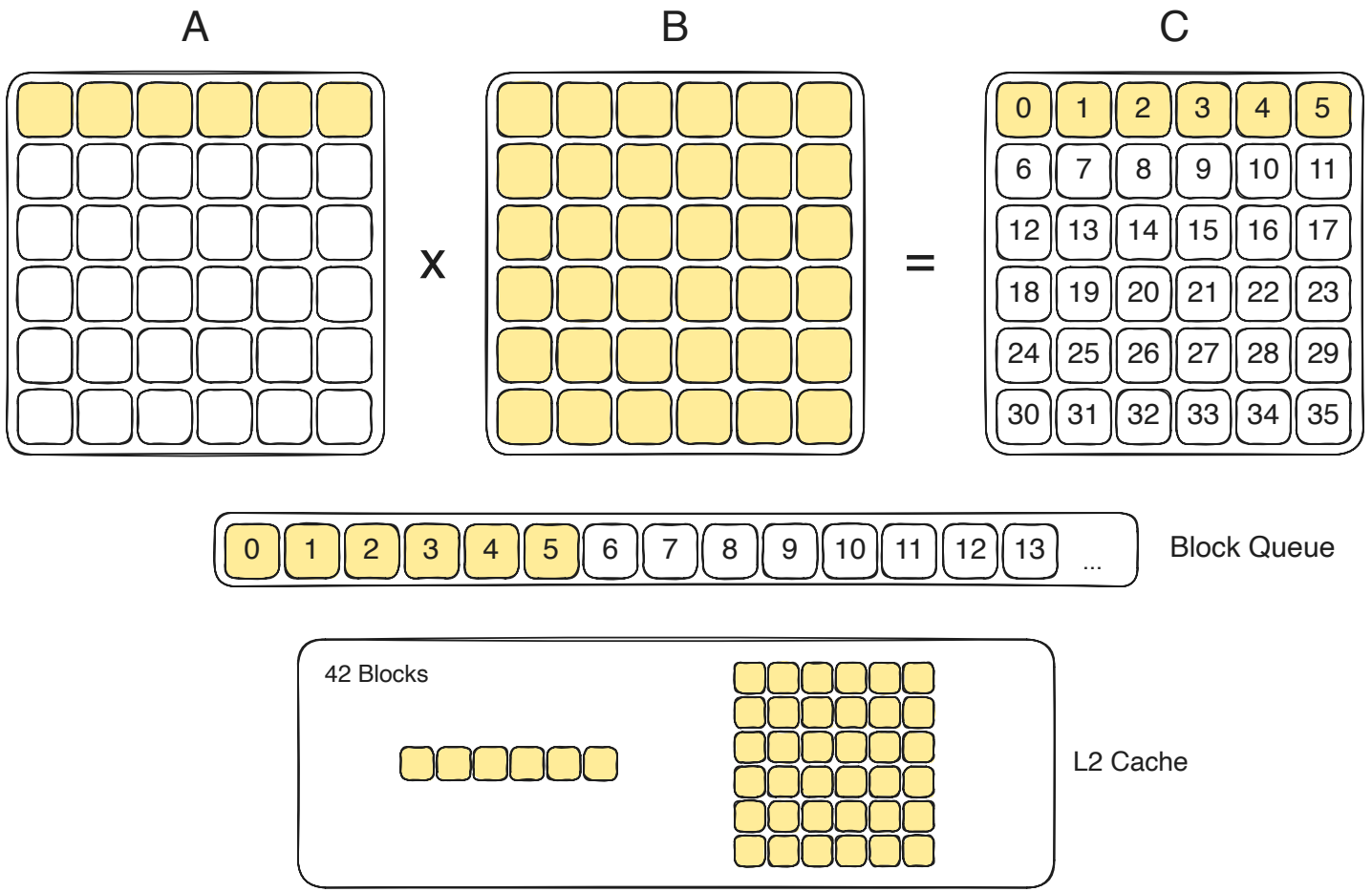
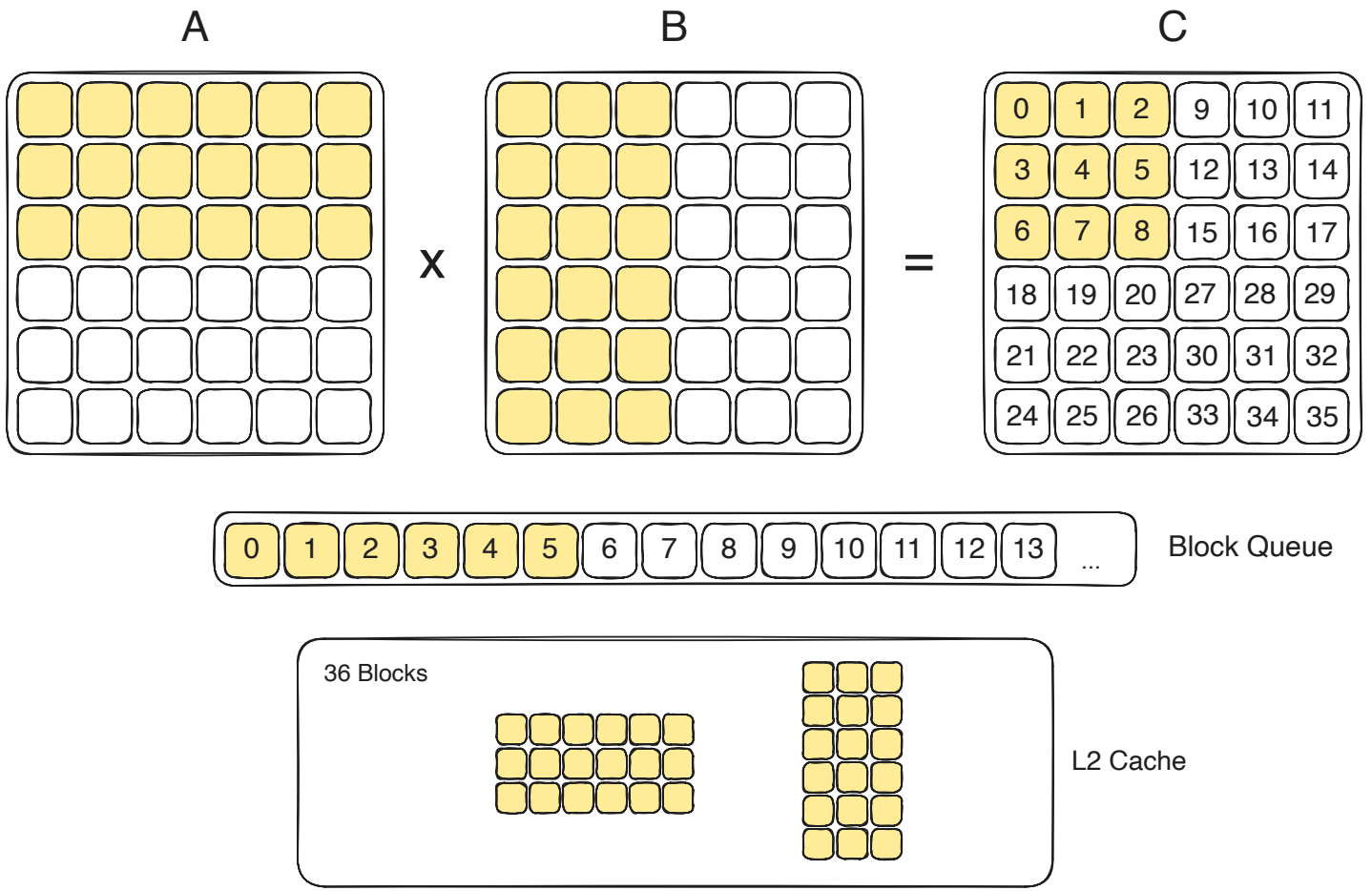


Thread Block Swizzling



Summary: Thread Block Assignment

- **In theory**, blocks can be scheduled in any order
- **In practice**, there is a strong bias towards scheduling blocks in order of their block IDs
- Naïve row-major block ID assignment can lead to poor L2 cache reuse
- **Thread block swizzling** can improve cache reuse and performance
  - Used to maximize **FLOPs / Bytes fetched from HBM**